

Problem

A constant current of 3 A for 4 hours is required to charge an automotive battery. If the terminal voltage is $10 + t/2$ V, where t is in hours,

(a) how much charge is transported as a result of the charging?

(b) how much energy is expended?

(c) how much does the charging cost? Assume electricity costs 9 cents/kWh.

Step-by-step solution

Step 1 of 4

Given

$$I = 3 \text{ A}$$

$$t = 4 \text{ hours}$$

$$t = 4 \times 60 \times 60 \text{ sec}$$

$$v = 10 + \frac{t}{2} \text{ V where } t \text{ is in hours}$$

Step 2 of 4

(a)

Charge = Current $\times$ Time

$$\text{Charge} = 3 \times 4 \times 60 \times 60$$

$$\text{Charge} = 12 \times 3600$$

$$\text{Charge} = 43200 \text{ C}$$

$$\text{Charge} = 43.2 \text{ kC}$$

Step 3 of 4

(b)

$$\text{Energy } E = \int_0^T p dt$$

$$E = \int_0^T vi dt$$

$$E = \int_0^T 3 \times \left(10 + \frac{t}{2 \times 3600} \right) dt \text{ (here we have to change the time into seconds)}$$

$$E = 3 \times \left[10t + \frac{t^2}{4 \times 3600} \right]_0^{4 \times 3600}$$

$$E = 3 \left[10(4 \times 3600) + \frac{(4 \times 3600)^2}{4 \times 3600} \right]$$

$$E = 3 \times 158400$$

$$E = 475.2 \text{ kJ}$$

(c)

Power is given by

$$P = \frac{475.2}{3600} \text{ kWh}$$

$$\text{Cost} = \left(\frac{475.2}{3600} \text{ kWh} \right) \times (9 \text{ cents/kWh})$$

$$\text{Cost} = 1.188 \text{ cents}$$